

4.00 USD

1. Sentinel Smart Valve Actuator (Model S-WFC01)

- *Based on Ecowitt WFC01*
- Smart water timer valve designed to mitigate localized wildfire threat by auto-wetting property grounds when dry soil moisture thresholds trigger parametric alarm states.
- **Price:** \$79.00 USD

2. Automated Network Onboarding Workflow

The Shopify purchase acts as the initial transaction in the node onboarding lifecycle.

```
``mermaid
sequenceDiagram
    autonumber
    actor User as Policyholder / Buyer
    participant Shop as Shopify Storefront
    participant Webhook as Shopify Webhook Worker
    participant API as Chainmail API Server
    participant DB as Firestore Database
    participant Device as IoT Sentinel Device

    User->>Shop: Purchase Sentinel Kit (S-100 + S-700)
    Shop-->>User: Order Confirmation & Shipping Label Generated
    Shop->>Webhook: Event: order/paid (Payload contains email, address)
    Webhook->>API: POST /api/onboarding/provision
    activate API
```

```
API→API: Generate secure device key & pre-provision propertyId

API→DB: Write property & pre-provisioned sensors as PENDING_ACTIVATION

API-->>Webhook: Provisioning Success

deactivate API

Webhook-->>User: Onboarding Email with Device Credentials & Claim QR Code

User→Device: Mount hardware & connect to home Wi-Fi

Device→API: POST /api/ingest/ecowitt (First Telemetry Ping)

activate API

API→API: Authenticate device with secure pre-provisioned key

API→DB: Update status to VERIFIED_NODE & Activate Parametric Coverage

API-->>Device: Acknowledge (HTTP 200)

deactivate API

API→User: App Push Notification: "Node Active. Coverage Live!"

、

---
```

3. API Provisioning Endpoint Specifications

When a checkout completes, Shopify sends an order/paid webhook to our backend API server:

Endpoint: `POST /api/onboarding/provision`

Access Control: HMAC Signature verified using Shopify API Webhook Shared Secret.

Request Payload (from Shopify):

```
`json
{
  "order_id": 987654321,
```

```
"email": "customer@example.com",

"shipping_address": {

  "address1": "12665 Julington Oaks Dr",

  "city": "Jacksonville",

  "province": "FL",

  "zip": "32223",

  "country": "US"

},

"line_items": [

{

  "sku": "SENTINEL-KIT-A",

  "quantity": 1,

  "name": "Sentinel Gateway & Wittboy Array"

}

]

}
```

Provisioning Logic:

1. **Address Validation:** Call Google Address Validation API to verify CASS status and retrieve exact rooftop coordinates (lat / lng).
2. **Database Record Creation:**
 - Create a new document in /properties:

```
`typescript
```

```
{
```

```
ownerEmail: "customer@example.com",

address: "12665 Julington Oaks Dr, Jacksonville, FL 32223",

lat: 30.1421936,

lng: -81.6157784,

riskScore: 72.4, // Calculated by Underwriting Agent based on coordinates

status: "PENDING_ACTIVATION",

createdAt: serverTimestamp()

}
```

- Create a document in `/properties/{propertyId}/sensors/Anemometry`:

```
`typescript

{

  type: "Anemometry",

  verificationStatus: "UNVERIFIED",

  hardwareType: "Ecowitt",

  value: 0

}
```

1. **Secret Generation:** Generate a unique password/secret token for the gateway and email it as a QR code to the user.

4. UI/UX Storefront Integration

The Shopify storefront theme will match Chainmail's core brand design system: **Cobalt & Gold / Slate & Gold**.

Theme Design System

- Primary Background: #030712 (Slate Black)
- Secondary Cards: rgba(17, 24, 39, 0.7) (Translucent Dark Blue)
- Accents: #D4AF37 (Metallic Gold)
- Fonts: Outfit (Headings) and Inter (Body)

Custom Liquid Elements

- Network Node Counter: A live ticker in the header showing: ACTIVE_NODES: 1,482 | TOTAL_COVERAGE_LOCKED: \$34,891,000`.
- Map Twin Preview: An interactive WebGL globe on the landing page showing glowing gold dots representing active coverage sensors in real time.
- onboarding-tab widget: Built-in section on the product page showing the simple "Plug, Scan, Insure" 3-step setup guide.

This playbook outlines the protocols for our internal sandbox testing phase. It defines how test results are tabulated, how errors are logged and escalated, how role-specific checkmarks are assigned to job codes, how users interact with their agentic co-pilots, and how we assign those agents.

1. How Internal Testing Results Are Tabulated

Testing results are tracked and aggregated across three distinct layers, providing real-time auditability:

```
graph TD
    mermaid
```

graph TD

```
A[Test Execution] → B[Layer 1: Central APM / Sentry Logs]
A → C[Layer 2: Local Firestore Test Ledger]
A → D[Layer 3: Intranet Dashboard Widgets]
B → B1[Raw Error Ingestion & X-Request-ID Tracking]
C → C1[Automated Script Runs & MAC Activation Matches]
D → D1[Operations Timeline & Full-Stack Audit Widget Updates]
```

1. **Centralized APM (Application Performance Monitoring) Logging:** Every transaction, API call, geocoding request, and claims state evaluation is tagged with a unique request header (X-Request-ID). If a test fails or encounters a timeout, Sentry routes a structured JSON log directly to our monitoring stack.
2. **Firestore Test Ledger (/test_runs):** A test coordinator script records automated test runs, capturing inputs, expected outputs, execution speeds, and pass/fail statuses.
3. **Visual Intranet Widgets (Consolidated View):**
 - [intranet_audit_widget.html]
(file:///C:/Users/greg_/OneDrive/Documents/Antigravity%20Save%20Folder/Chainmail-Intranet-Widgets/intranet_audit_widget.html): Displays the high-level health (0-100%) of all 13 full-stack architectural layers.
 - [intranet_operations_timeline.html]
(file:///C:/Users/greg_/OneDrive/Documents/Antigravity%20Save%20Folder/Chainmail-Intranet-Widgets/intranet_operations_timeline.html): Displays a date-stamped log showing the specific ticket or test-case ID, description, and status (DONE, TO DO, QUEUED, PENDING_DVN).

2. What Happens When the System Hits an Error

To preserve capital and protect the network from incorrect state changes, we enforce strict fail-safe protocols:

1. **Immediate Hard-Halt and State Exceptions:**
 - ***Geospatial Coordinate Drift:** If coordinates returned by geocoding API drift beyond the 0.0001-degree threshold from static imagery baselines, the system halts processing and raises a `GeospatialDriftException`.

- ***Confidence Intervals:** If multi-factor confidence calculations (evaluating local permits, municipality maps, and historical satellite tracks) drop below 95%, the automated underwriting engine halts, pausing active premium quotes.

1. The Adverse Non-Automation Rule (Human-in-the-Loop Handoff):

- Under Florida OIR and New York DFS Part 500 regulations, Agentic AI is strictly prohibited from executing autonomous adverse claims actions (denials or cancellations) without licensed human oversight.
- When a validation error or threshold breach occurs, the Agent cannot deny the file. Instead, the backend halts automated processing, marks the document status as `PENDING_ARBITRATION`, and pushes a alert panel containing raw JSON payloads directly to the human Integrity Officer's console.

1. **Backup Data Routing:** If a physical weather station (IoT gateway) goes offline or suffers telemetry packet loss, the Ingestion Agent registers a disconnect, switches to backup government weather feeds (NOAA/NWS) to check regional parametric bounds, and logs a recovery notice.

3. Job Code Testing Matrix

To maximize test coverage, coworkers will evaluate the platform features corresponding to their operational domains and job descriptions:

A. Executive Level (CEO / Strategic Oversight)

- **Target Domain:** Domain 4 (Treasury & Cryptographic Governance).
- **Testing Focus:**
- Verify the 3-of-5 multi-signature threshold sequence for capital reserve tranches.
- Verify Bitcoin treasury hedging triggers and check the USD-denominated digital reserve asset conversion ledgers.
- Review high-level Intranet budget burn widgets for accuracy.

B. Operations & Product Management (COO / PMs)

- **Target Domain:** Domain 1 (Operational Lifecycle) and Domain 5 (IoT Ingestion).
- **Testing Focus:**
- Simulate Shopify kit checkouts to verify the automated `/api/onboarding/provision` webhook post routes.
- Test device MAC address pre-provisioning bindings in the Firestore sensors collections.
- Trigger high-frequency simulated telemetry inputs to verify "verified node" transitions.

C. Legal, Risk & Compliance (CRO / Regulatory Officers)

- **Target Domain:** Domain 6 (Regulatory Compliance & Zero-Knowledge Privacy).
- **Testing Focus:**

- Verify Zero-Knowledge proving outputs (π) to ensure no personally identifiable information (PII) or raw geocoded coordinates leak on public ledgers.
- Test state compliance automated reports (e.g., FL OIR-B1-2222 metrics for timing constraints).
- Audit the human-override adjuster panel to confirm the Adverse Non-Automation rule holds.

D. Underwriting & Actuarial (Lead Actuaries / Underwriters)

- Target Domain: Domain 2 (The Quant Brain & Parametric Pricing).
- Testing Focus:
- Inject extreme coordinate drift inputs to confirm geocoding halt rules trigger.
- Verify underwriting risk pricing results across different hurricane zones.
- Stress-test confidence interval thresholds.

4. Agent Assignments & Interaction Mechanics

Every role within the Chainmail ecosystem is assigned an Agentic Co-Pilot. Users interact with their agents through unified intranet components.

`

+-----+

INTRANET WIDGET LAYOUT	
[Operations Timeline] [Floating AI Knowledge Companion]	
* OPS-257 (Done)	Connected: Underwriting_Agent_01
[Role Doctrine: Actuary]	User: "Check risk at 12665 Oaks"
Access: Read/Write Models	Agent: "Geocoding coordinates..."
[Sidecar Action Panel]	[Send prompt] [Edit system]

+-----+

How We Assign Agents

1. **Authentication Mapping (ABAC):** When a user logs in via Google OAuth, the system inspects their authenticated email against the `/users/{uid}` database registry.
2. **Role Provisioning:** Firestore assigns the user a corresponding role claiming tag.
3. **Agent Binding:** Based on the claiming tag, the user is paired with a specific backend Agent model:

- Manager, AI Implementation $\rightarrow$ Antigravity (Core Agent)
- Lead Actuary AI $\rightarrow$ Chief Underwriting Agent
- Integrity Officer / adjuster $\rightarrow$ Claims Verification & Ingestion Agent

How Users Interact with Their Agents

- **Intranet Floating AI Chat Widget:** A persistent, gold-accented chat window is embedded in the Intranet workspace. The user chats directly with their assigned agentic helper, allowing them to query database logs, audit codebases, or run tasks.
- **Action Sidecars:** Inside the main dashboard, the Agent populates a sidecar pane with action cards (e.g., "Draft Claims Invoice Ready for Approval"). The human clicks "Approve/Reject" to complete the execution block.
- **Doctrine Access:** The [Operational Guidelines Widget] (file:///C:/Users/greg/OneDrive/Documents/Antigravity%20Save%20Folder/Chainmail-Intranet-Widgets/intranet_roles_widget.html) links each role directly to their corresponding operational manuals and prompt structures, ensuring complete alignment.

5. Phased Sandbox Rollout Proposal

If you are not comfortable opening the system to all users immediately, we recommend a Phased Sandbox Testing Framework to isolate variables and maintain complete control:

`

[Phase A: Greg Single-User] ---> [Phase B: Co-Founder Audit] ---> [Phase C: Department Sandbox] ---> [Phase D: Full Team Launch]

`

- ****Phase A: Single-User Validation (Greg Only) – *Current****
- Only Greg's credentials have access rules in Firestore. No other users can log in or view databases. All live geocoding tests are run directly by Greg.
- **Phase B: Co-Founder Audit (2 Users)**
- Assign access specifically to your co-founder. Together, run sandbox transactions (simulated Shopify checkout, simulated claims trigger) to verify payments and multi-signature payouts.

- **Phase C: Phased Department Sandboxes (5 Users)**
 - Enable access for the Lead Actuary and CRO. Restrict database access to a dedicated /sandbox` collection. Allow them to test underwritings and regulatory audits without write access to production database folders.
- **Phase D: Full Team Launch**
 - Promote verified roles to standard production scopes, activate Google OAuth global logins, and monitor system integrations.

,500 payout cap) just so we can get their sensor data and help protect the community.

The Claims Adjuster Agent: When a disaster hits, this AI looks at satellite photos to verify the damage. If it's confident, it pays the customer instantly. If it isn't sure, it sends the claim to a human to review—it is strictly programmed **never** to automatically deny a claim.

The Vigilance Oracle Agent: This AI constantly watches the weather channel. If it sees a Category 5 hurricane coming, it flags the map and readies our bank accounts to pay people before they even ask.

The Quant Brain Actuary Agent: This AI watches our bank account. If we take on too much risk in one state, it automatically buys "reinsurance" (backup insurance for us) to keep the company financially safe.

Your Day-to-Day Responsibilities

As a Product Owner, you will:

- **Monitor the Vision:** Look at the Enterprise Dashboard. Are the AI agents making decisions that align with the BallPark Initiative?
- **Prioritize the Backlog:** Work with the Product Manager to decide which neighborhood of our "city" needs improvement next.
- **Understand the Costs:** The Enterprise dashboard shows you exactly how many fractions of a cent the AI is spending to make decisions. You balance the cost of the AI against the value it provides.

You don't need to know how to code the AI. You just need to know what the AI **should** be doing to help our customers. Welcome to the team!

,500."*

Here is how you handle it:

- Explain that because their house is in an **Extreme Risk Zone** (like right on the beach), our Underwriting AI cannot offer them half a million dollars right now.
- However, explain that we **did not reject them**. We offered them the **IoT Baseline Policy**. This gives them ,500 of emergency cash, and in return, they install a weather sensor on their house. This helps us gather data to protect the rest of their neighborhood!

Need Help?

If the screen turns white, or the map stops loading, do not panic! Just take a screenshot and send it to your Product Manager with the Jira ticket label UI-BUG.

Welcome to Chainmail! We are glad you're here.

5). Require D-U-N-S number for corporate verification. **TestFlight / Internal Testing** T-4 Weeks Release to internal stakeholders via Apple TestFlight and Google Play Internal Testing. **Store Submission** T-2 Weeks Submit to Apple App Review and Google Play App Review. (Buffer time for initial rejection and resubmission). **Public Launch** T-0 App is live, indexed, and available for download.

2. Critical Apple App Store Requirements (iOS)

Apple is notoriously strict regarding financial and insurance applications. To pass the App Review Board, we must guarantee the following:

WARNING

digital settlement & Financial Regulations (Guideline 3.1.5)

Because Chainmail utilizes automated execution rules on Secure L2 Ledger L3 for payouts using \$USD-denominated digital settlement assets, the app **must** be submitted by a legal entity (not an individual developer) that is registered as a financial institution or has the appropriate licenses in the regions it operates.

- **Action Item:** Ensure our corporate D-U-N-S number is used for the Apple Developer Account, and our legal entity name exactly matches the app publisher name.

IMPORTANT

Data Privacy & Account Deletion (Guideline 5.1.1)

Apple absolutely mandates that any app allowing account creation **must** also provide an in-app option to completely delete the account and all associated data.

- **Action Item:** Build an automated "Delete Account" API route in the Vercel backend.

- **App Tracking Transparency (ATT):** If we use analytics that track across third-party apps, we must implement the ATT prompt.
- **Minimum Functionality:** The app cannot just be a "wrapped website" (Guideline 4.2). It must utilize native iOS features (e.g., Push Notifications, FaceID/TouchID for secure login, local caching).

3. Critical Google Play Requirements (Android)

Google Play has recently tightened restrictions on testing and financial apps.

WARNING

20-Tester Rule

New Google Play developer accounts must run a closed test with a minimum of 20 opted-in testers for at least 14 continuous days before you can apply for production access.

- **Action Item:** We must recruit 20 beta testers internally or via the Vigilance Grid network well before launch.

IMPORTANT

Financial Features Policy

Google requires a prominent "Financial Features Declaration" during submission. If the app facilitates insurance or digital settlement payouts, we must provide links to our regulatory compliance documentation and operating licenses in our Google Play Console declaration.

- **Data Safety Section:** We must accurately declare exactly what data is collected (e.g., location data from Vigilance Grid IoT integration, financial info) and whether it is encrypted in transit.

4. Next Steps for the Development Team

To transition our current Vercel web app into a compliant Native App:

1. **Architecture:** We must decide whether to wrap the Vercel app using a framework like Capacitor/Ionic, or rebuild the frontend natively using React Native.
2. **Privacy Policy URL:** A live, hosted Privacy Policy must be created on the public-facing website.
3. **Support URL:** A dedicated support and contact page must be established for user inquiries.

0/mo) post-funding for team access. **Database & Scaling** GCP Firestore / Blaze Plan 00.00 Scales dynamically; covers production compute and database queries. **Object Storage (Media/PII)** Cloudflare R2 \$0.00 Selected specifically to eliminate AWS/GCP data egress fees. **Geospatial &**

Convective Mapping XWeather API & Vexcel Imagery \$750.00 Commercial subscription for severe convective storm mapping and high-res post-storm imagery. **Core AI Routing & HITL Verification** Google Gemini 1.5 Flash & 3.1 Pro \$65.00 Production Gemini API tokens for IoT routing and human-in-the-loop validation. **Claims Fraud Audit** ISO Verisk ClaimSearch 50.00 Active double-claim checking API subscription. **Secure Registry & Smart Contract execution** Secure L2 Ledger Mainnet 5.00 Sponsored gas for transaction settlement. **Version Control** GitHub (Free Tier) \$0.00 Upgrade to GitHub Team (\$4/user) when engineering team expands. **Executive Compensation** COO Phase 1 (1099) \$0.00 (Equity/\$RMOR only) Transition to Full-Time Phase 1 (Salary + 8% Match) post-funding. **Total Monthly Burn** ,090.00

Future Capital Expenditure (Post-Funding)

Once institutional seed funding is secured, the following immediate upgrades will be unlocked to mature the ecosystem:

1. **CFAN Proprietary Licensing:** We will deprecate our reliance on open-source NWS/NCEP data and execute a commercial licensing agreement with the **Climate Forecast Applications Network (CFAN)**. This will drop our 5-day track error to ~131 miles, providing institutional credibility to reinsurers. *(Estimated: Negotiated Enterprise Contract)*
2. **Amazon IP Accelerator:** We will abandon the standard USPTO pending application process and utilize Amazon's vetted legal firms to fast-track our Sentinel IoT hardware Trademark and Brand Registry. *(Estimated: ,500 - \$3,000)*
3. **DVN Node Infrastructure:** Funding the initial liquidity pools on Secure L2 Ledger to incentivize legacy Third-Party Vendors (TPVs) to transition into distributed Validator Network (DVN) node operators.